

Golden Ratio φ , e , π and Fine Structure Constant α : Collapse Breathing Proportions

Haobo Ma^{1*}, Wen Niu¹

¹AELF PTE LTD., 8 Marina Blvd, Singapore 018981

*Correspondence to: auric@aelf.io

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Abstract

This paper investigates the intrinsic mathematical relationships between four fundamental constants: the Golden Ratio (φ), Euler's number (e), Pi (π), and the Fine Structure Constant (α). Using the Universe Ontology framework based on FLIP-XOR-SHIFT operations, we demonstrate that these seemingly disparate constants form a unified system of proportions that governs fundamental reality structures. We introduce the concept of "collapse breathing proportions" to describe how these constants interrelate through information field dynamics. Our derivations reveal that the fine structure constant can be expressed through an exact formula involving φ , e , and π , providing a new theoretical foundation for understanding this mysterious dimensionless constant. Numerical simulations validate these relationships to high precision. This work suggests that the fundamental constants of nature are manifestations of underlying information-theoretic principles rather than arbitrary values, with significant implications for our understanding of physical reality.

Keywords: Fine structure constant, Golden ratio, Mathematical constants, Universe Ontology, Information theory, FLIP-XOR-SHIFT operations

1 Introduction

2 Theoretical Framework

3 Mathematical Formulation

4 Results and Analysis

5 Discussion

6 Conclusion

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